

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-VI (NEW) EXAMINATION – SUMMER 2024****Subject Code: 3161613****Date:20-05-2024****Subject Name: Data Analysis and Visualization****Time: 10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Describe Descriptive and Inferential statistics.	03
	(b) What do you mean by probability? Explain characteristics of probability.	04
	(c) Explain “The Big Three” in Data Visualization in detail.	07
Q.2	(a) Differentiate MultinomialNB, BernoulliNB and GaussianNB.	03
	(b) Explain Reducing Dimensionality with factor analysis.	04
	(c) Explain methods for calculating correlation.	07
	OR	
	(c) Explain methods of Regression in detail.	07
Q.3	(a) Differentiate supervised and unsupervised learning.	03
	(b) Explain how to categorize data using random forest algorithm.	04
	(c) Explain the working of k- nearest neighbor algorithm.	07
	OR	
Q.3	(a) Describe Normal, Binomial and Categorical distributions.	03
	(b) Explain how to categorize data using decision tree algorithm.	04
	(c) Explain Hierarchical clustering algorithm.	07
Q.4	(a) List various Open-Source Web-Based Data Visualization Platforms. Explain any one of them.	03
	(b) Which are the “Standard Charts”? Explain in brief.	04
	(c) What is D3.js? Explain .selection, .selectAll and .data() in brief.	07
	OR	
Q.4	(a) List different types of Comparative graphics for data visualization and explain any one of them.	03
	(b) What are the significance of SVG and DOM in D3.js.	04
	(c) Write a short note on types of target audience and suitable graphics for data visualization.	07
Q.5	(a) Illustrate the layer overlay analysis in GIS with diagram.	03
	(b) What is Infographics? Discuss any one of following tool in brief. (1) Infogr.am (2) Piktochart	04
	(c) List and explain steps to focusing on the audience while designing dashboard in brief.	07
	OR	
Q.5	(a) Explain buffering and proximity analysis in GIS.	03
	(b) Explain AARRR of the e-commerce sales funnel.	04
	(c) What is web analytics? Explain the use of various applications for web analytics.	07
